Math 110 Hard Mock Midterm 2

50 minutes 4 questions 40 points

Set a 50-minute timer. Give concise reasoning with each answer. You may quote results from the notes or text if you state them. $\mathcal{P}_n(\mathbb{R})$ is the space of real polynomials of degree at most n. All bases are ordered.

1. Two subspaces of matrices (8 points) In $V = \mathbb{R}^{2,2}$ let

$$U = \left\{A : A\begin{pmatrix}1\\1\end{pmatrix} = 0\right\}, \qquad W = \{A : A^T = A\}.$$

a) (3) Find $\dim U$, $\dim W$, and a basis of $U \cap W$.

b) (3) Is $V = U + W$? Is the sum direct? Prove both answers.

c) (2) Find a subspace $W' \subseteq W$ with $V = U \oplus W'$, and prove it works.

2. Factoring a map (10 points) Let V, W, X be finite-dimensional, and let $S\colon V \to X$ and $T\colon V \to W$ be linear.

a) (2) Prove: if $T = RS$ for some $R \in \mathcal{L}(X, W)$, then $\operatorname{null} S \subseteq \operatorname{null} T$.

b) (3) Now assume $\operatorname{null} S \subseteq \operatorname{null} T$. For $y \in \operatorname{range} S$, define $R_0(y) = Tv$, where v is any vector with $Sv = y$. Prove that $R_0\colon \operatorname{range} S \to W$ is well defined and linear.

c) (3) Prove that there is $R \in \mathcal{L}(X, W)$ with $T = RS$.

d) (2) Let $S\colon \mathcal{P}_3(\mathbb{R}) \to \mathcal{P}_1(\mathbb{R})$ be $S(p) = p''$. For $T(p) = p'(1) - p'(0)$, find an explicit R with $T = RS$. For $T(p) = p(0)$, prove no such R exists.

3. Nonstandard bases on both sides (10 points) Define $T\colon \mathcal{P}_2(\mathbb{R}) \to \mathbb{R}^3$ by

$$T(p) = \big(p(1),\ p'(1),\ p(2)\big).$$

Use the basis $B = \big(1,\ x-1,\ (x-1)^2\big)$ of the domain and $C = \big((1,0,0),(1,1,0),(1,1,1)\big)$ of the codomain.

a) (4) Find $\mathcal{M}(T)$ with respect to B and C.

b) (3) Prove that T is an isomorphism without using the matrix.

c) (3) Find the $p \in \mathcal{P}_2(\mathbb{R})$ with $T(p) = (1,1,1)$, and write it in the standard basis.

4. Preimages through a quotient (12 points) Let V and W be finite-dimensional, $T \in \mathcal{L}(V, W)$, and U a subspace of W. Define

$$S\colon V \longrightarrow W/U, \qquad S(v) = Tv + U.$$

a) (3) Prove that S is linear and that $\operatorname{null} S = T^{-1}(U)$.

b) (3) Prove that $\dim \operatorname{range} S = \dim(\operatorname{range} T + U) - \dim U$.

c) (3) Deduce $\dim T^{-1}(U) = \dim \operatorname{null} T + \dim(U \cap \operatorname{range} T)$.

d) (3) Let $D\colon \mathcal{P}_4(\mathbb{R}) \to \mathcal{P}_4(\mathbb{R})$ be differentiation and $U = \operatorname{span}(x^3, x^4)$. Use (c) to find $\dim D^{-1}(U)$, then give a basis of $D^{-1}(U)$.